# Harish G

**Systems & Platform Operations | Production Support, Monitoring & Automation**

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## Profile

I stay with a system after it ships. On a public-facing telephony platform I own the call pipeline in production: fixed outbound routing, added reconnect handling for dropped Gemini 1006 sockets, and traced a Docker TLS failure back to NAT hairpinning on a self-hosted domain, which read as an application bug until somebody followed the packet path. I build the observability in rather than bolt it on, and write the runbook that says what to do at 2am. Software Developer at CortexCraft.ai.

## Core Skills

**Operations** 24x7 ownership of a live, public-facing call pipeline · root-cause analysis from symptom to packet path · runbooks, a hotfix flow and a rollback procedure · backend and provider migrations run without interrupting live traffic

**Monitoring** single-line structured JSON logging · `X-Request-ID` correlated end to end · per-response status and duration in milliseconds · Prometheus instrumentator on `/metrics` · health endpoints · a real-time WebSocket monitoring dashboard over a Windows desktop agent · append-only audit logs with TTL retention

**Platform** Linux hosts under Docker and Docker Compose · Jenkins, GitLab CI and GitHub Actions as the merge gate · containerised multi-service deploys · TLS, DNS and NAT fault isolation · configuration in the database, so a change is a config row and not a deploy

**Scripting** Python · Java · JavaScript / TypeScript · SQL · Celery workers and scheduled jobs · bulk CSV and Excel import / export jobs

**Databases** PostgreSQL with Alembic and Flyway versioned migrations · MongoDB · Redis (locks and queues) · Neo4j · credentials encrypted at rest

**Automation** External REST API that other products drive · webhook callback handlers · ticket-to-pull-request workflows over the Jira and GitHub APIs · 132 pytest files acting as the CI merge gate

## Experience

**Software Developer, CortexCraft.ai** Jan 2026 – Present

- Authored 589 of 626 commits on the voice-agent SaaS platform and 67 of 75 on its core engine, and own production reliability of the call pipeline it runs: fixed outbound routing, added reconnect handling for dropped Gemini 1006 sockets, and traced a Docker TLS failure to NAT hairpinning on a self-hosted domain.
- Lead author (172 of 192 commits) on the multilingual survey platform. Migrated it off a Node backend onto FastAPI and off one speech provider onto another behind a config-driven registry of 24 locales, both without interrupting surveys that were running at the time.
- Run the build and deploy path behind it: Jenkins and GitLab CI, Docker Compose, 132 pytest files gating every merge, and Alembic-migrated PostgreSQL treated as a release.
- Maintain integration workflows against two telephony vendors and several partner APIs: an external REST API other products call, webhook callback handlers, per-tenant credentials encrypted at rest, and an agent that turns Jira tickets into reviewed pull requests.

**Freelance Web & App Developer (Self-Employed)** 2023 – 2026

Delivered client systems end to end, then supported them: deployment, environment changes, upgrades and fixes, with direct client contact for triage and release sign-off.

## Selected Projects

**CortexCraft Voice Agent, production telephony platform** *Python, FastAPI, WebSocket, Docker, React*

- 589 of 626 commits. One platform behind several shipped products, so a fix to the call pipeline reaches all of them at once. Most of the real work was keeping it up: routing, socket reconnects and TLS, each traced to a cause rather than restarted past.

**KisanVoice AI, multilingual survey platform** *Python, FastAPI, MongoDB, Jenkins, Docker Compose*

- 172 of 192 commits. Operations built in rather than added later: every request carries an `X-Request-ID` that is forwarded or generated and echoed back, every log line is a single JSON object carrying that ID, each response is logged with its status and duration in milliseconds, and a Prometheus instrumentator exposes `/metrics`.
- Ships with a deployment runbook covering prerequisites, environment setup, a verification step and a symptom-to-fix section for the failures the team actually hit, plus a written release process with a hotfix flow and a rollback procedure. 24 configured locales come from a config-driven registry, so adding one is a config row rather than a deploy.

**LeadCall AI, multi-tenant outbound calling** *FastAPI, Celery, PostgreSQL, Redis, React*

- Queued the slow work instead of blocking on it: Celery dial and callback workers behind Redis dial locks, per-tenant credentials encrypted at rest, bulk Excel import and call-log export, and an append-only audit log of every operator action.

## Education & Ways of Working

**B.E., Information Science and Engineering**, Vivekananda Institute of Technology, Bangalore. One issue per branch, tests as the merge gate, a runbook entry for every failure seen twice.